ASSA ABLOY is the global leader in door opening solutions, dedicated to satisfying end-user needs for security, safety and convenience



DE6090AR Automatic Restraint System Introduction of the new product

DE6090AR Automatic Restraint



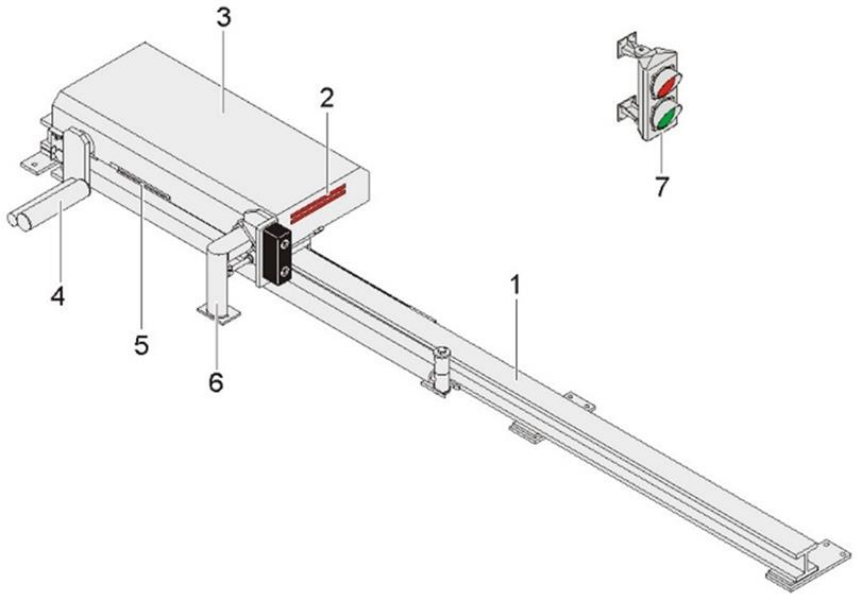
- Picture shows UK Version

DE6090AR Automatic Restraint

- Background
- Safety has become a strong business driver all over Europe
- Restraint Systems that used to a be niche product is becoming a standard in the building specifications mainly for the Manufacturing and D&L segments

- French statistics of 2017 shows the following picture:
  - Logistic is 10% of GDP
  - 2 Mio employees
  - 70 deadly accidents per year (partly during docking operations)
  - 36.000 injuries per year (partly during docking operations)

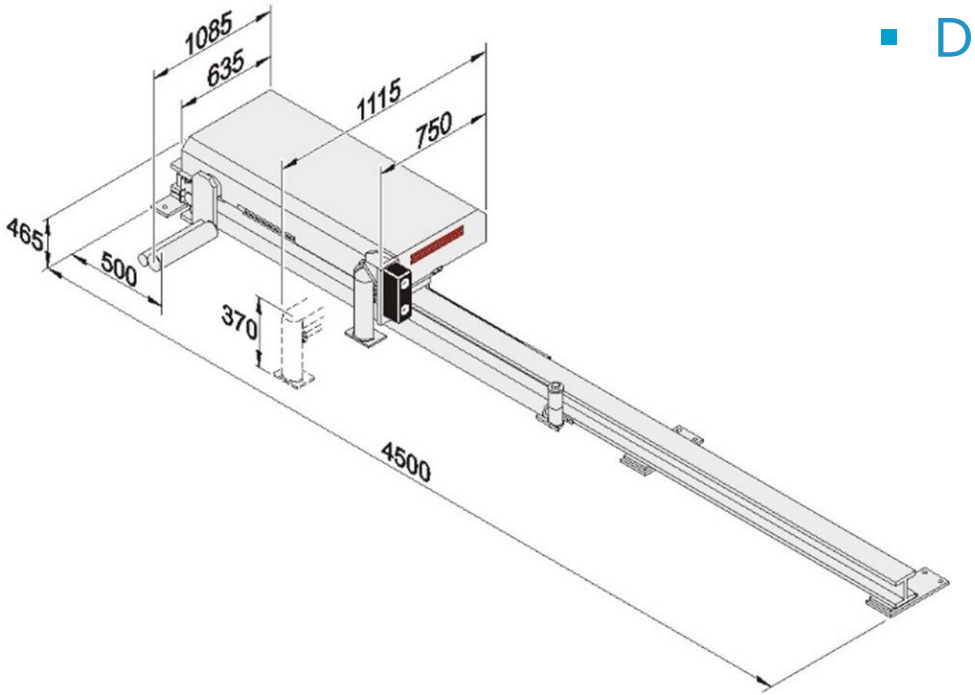
DE6090AR Automatic Restraint



- Overview

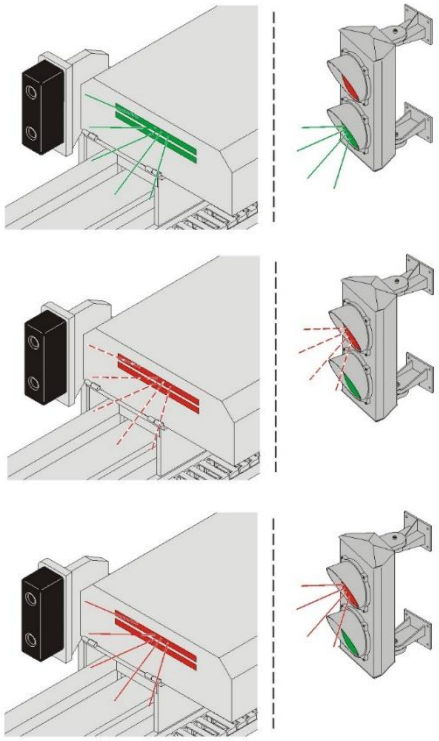
1. Guide rail
2. Integrated traffic light
3. Carriage
4. Detection arm
5. Outside LED light
6. Blocking arm
7. Additional outside traffic light

DE6090AR Automatic Restraint



- Dimensions

DE6090AR Automatic Restraint



- Outside Traffic Light / Truck Driver Instructions
- GREEN

  Truck can arrive or drive off

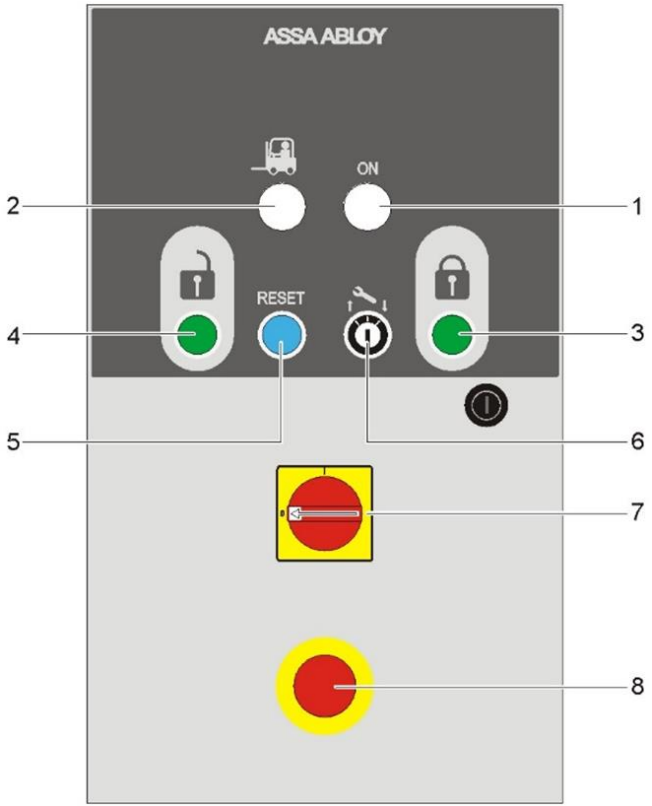
- RED BLINKING (and beeper alarm)

  Attention. The carriage is moving.

- RED

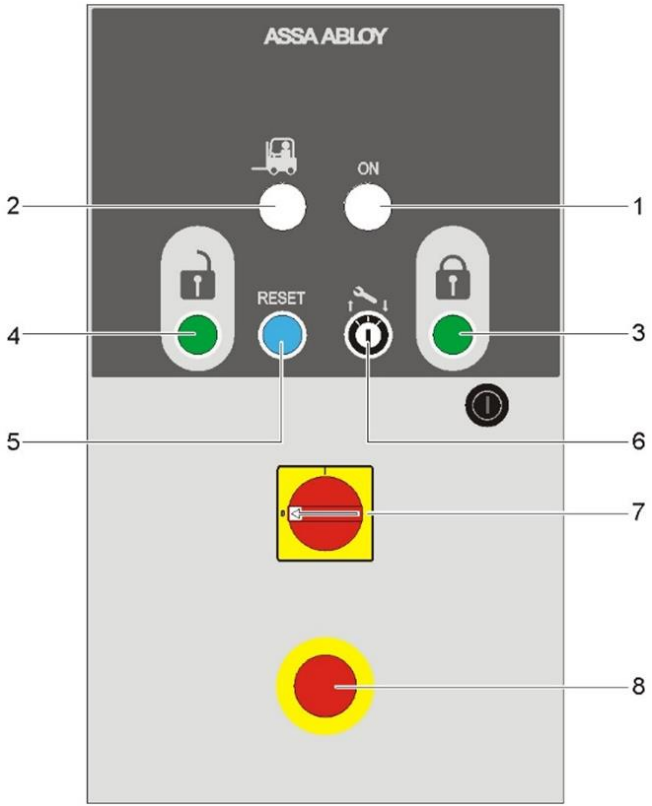
  Truck is blocked

DE6090AR Automatic Restraint



- Control Box

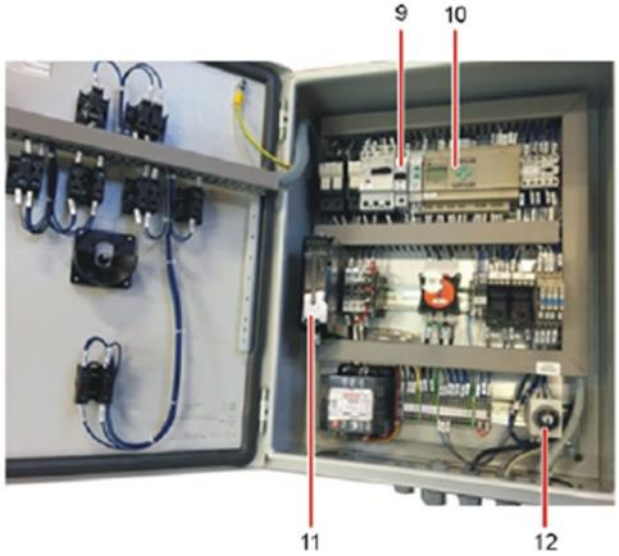
DE6090AR Automatic Restraint



Control Box

1. Signal white to indicate power ON.
2. Signal green to indicate leveler or sectional door is released (the truck is docked and blocked).
3. Push button to activate the restraint system to move forward and block the truck. The push button lamp is green when the restraint system is ready for use or when the optional truck detection sensor is activated.
4. Push button to release the restraint system to move back to the parking position.
5. Push button to RESET the restraint system.
   Signal blue to indicate an error in the system or when emergency push button has been activated.
6. Key switch for maintenance.
7. Mains Switch.
8. Emergency stop push button.

DE6090AR Automatic Restraint



Control Box

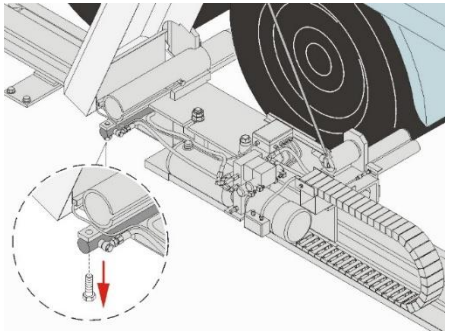
- 9. Switch for Summer/Winter mode
  In Winter mode the hydraulic unit will run in a closed circuit for one minute, every four hours. Recommended during winter time when the outside temperature constantly is below 0 degrees Celsius.
- 10. Automation with Display
- 11. Stabilized Power Supply
- 12. Key Switch to override the interlocking functions
  Use the docking station without the restraint system when a vehicle (e.g. de-mountable containers on legs) cannot be blocked by the restraint system.

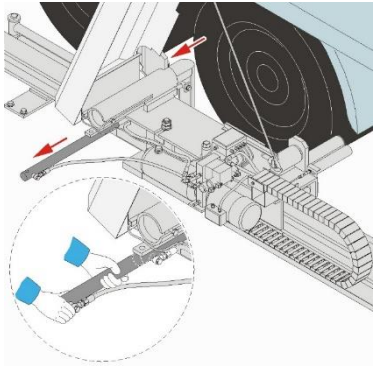
DE6090AR Automatic Restraint

- Standard
- Restraint force:200kN
- Motor cable 15 m
  (for leveler behind the door & Loadhouse)
- Interlocking sensor for Sectional Door and Dock Leveler
- Traffic light Integrated on the carriage
- Outside LED light with a brightness sensor, lights up when is dark.
- Override key switch that disconnects the interlocking functions.
- Winter mode switch
- Manual release of the blocking arm

- Options
- Additional outside Traffic Light
  (cable length 12 m)
- Sensor for Truck Detection
  (Dock-IN sensor)
- Parking guides

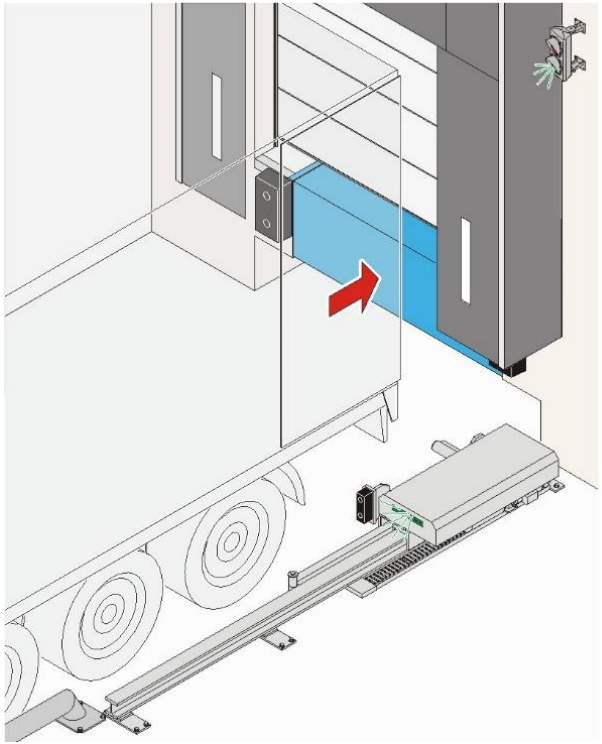
DE6090AR Automatic Restraint





- Manual Release of the Blocking Arm
- This function is a solution to avoid interruptions when
  - There is a failure in the system
  - There is an interruption of the power supply
  - There is too high pressure from the truck on the blocking arm due to wrong positioning of the truck

DE6090AR Automatic Restraint

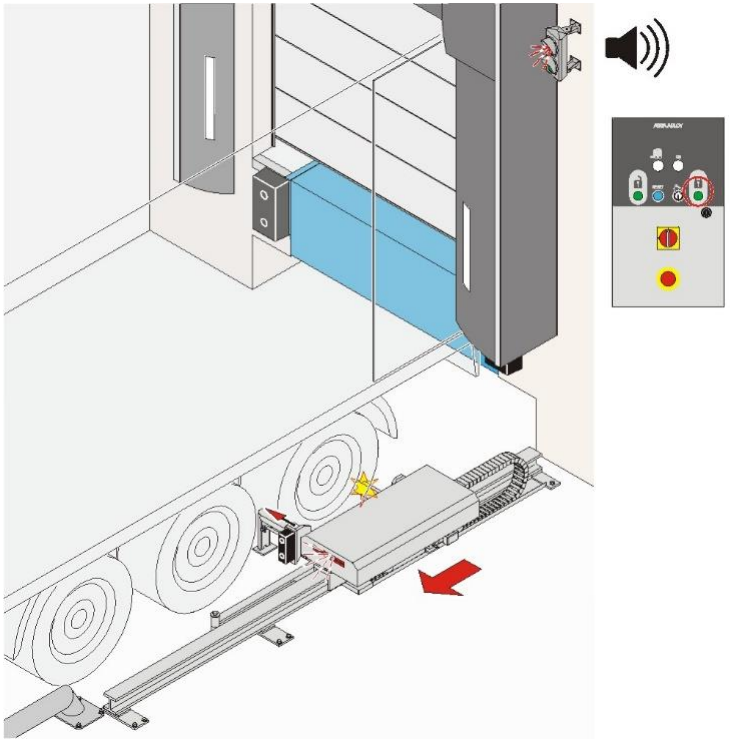


- Mode of operation

1. Ready for use condition

- GREEN
- The loading operation cannot be started.
- The restraint system is blocked for use
- The sectional door or the dock leveler is blocked for use.

DE6090AR Automatic Restraint

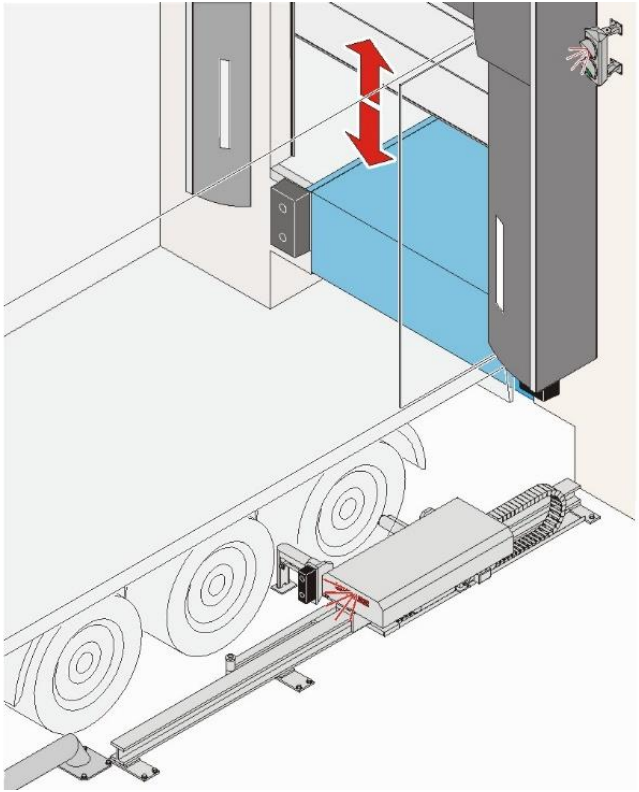


- Mode of operation

2. Activate the Restraint System

- Flashing RED and Beeper Alarm during the movement.
- Detection arm contacts the rear truck wheel
- Blocking arm blocks the truck
- Then RED
- The sectional door or the dock leveler is released for use.

DE6090AR Automatic Restraint



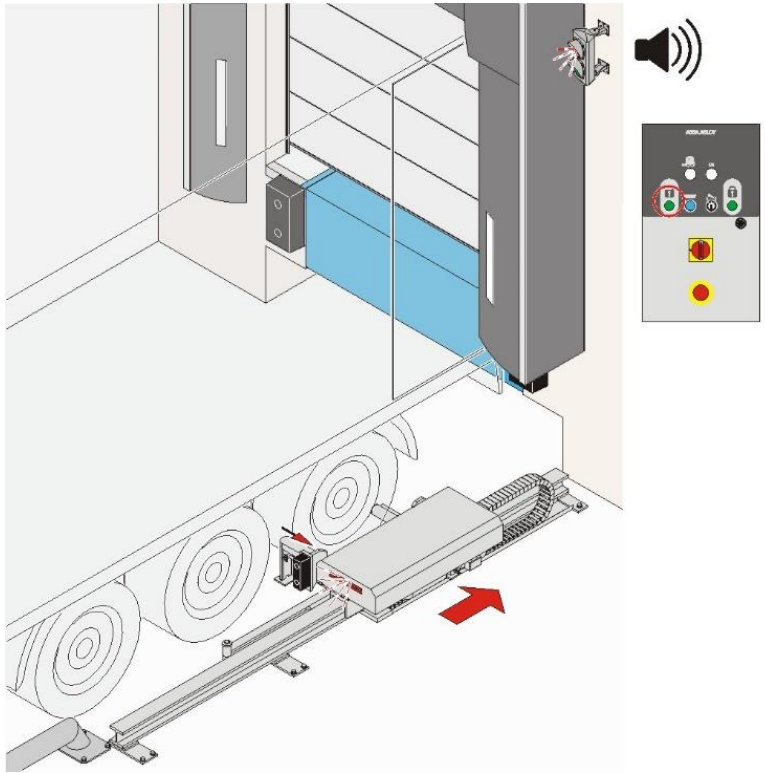


- Mode of operation

3. Loading Operation

- RED
- Loading operation
- Open Sectional Door and connect the dock leveler to the truck
- The truck cannot roll away or drive off!
- The Restraint System is blocked for use

DE6090AR Automatic Restraint

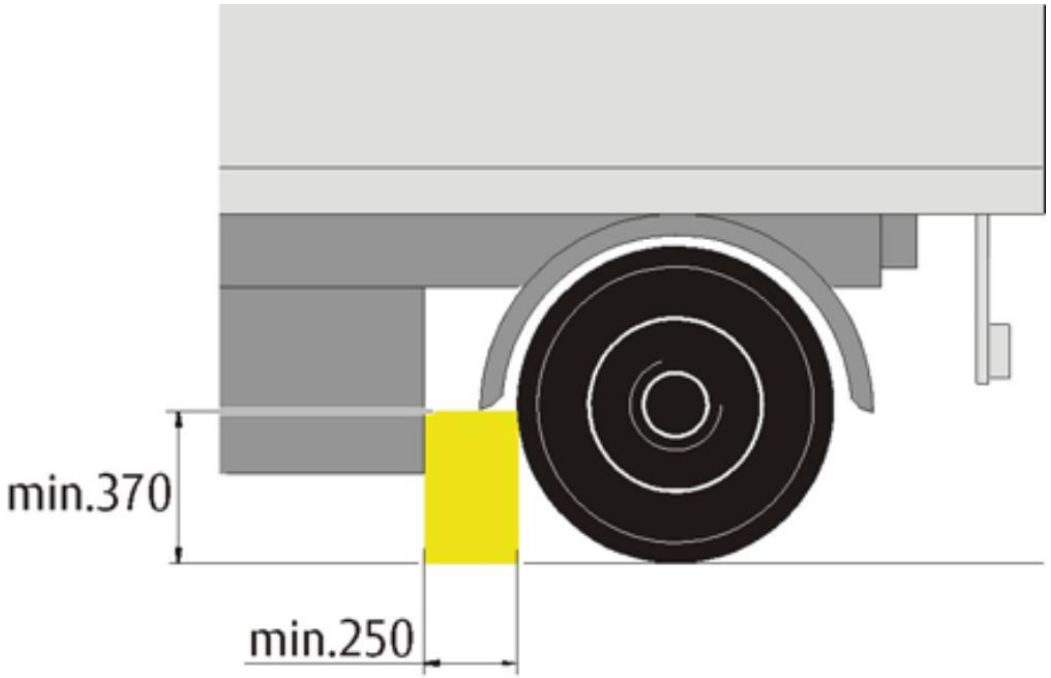


- Mode of operation

4. Release the Restraint System

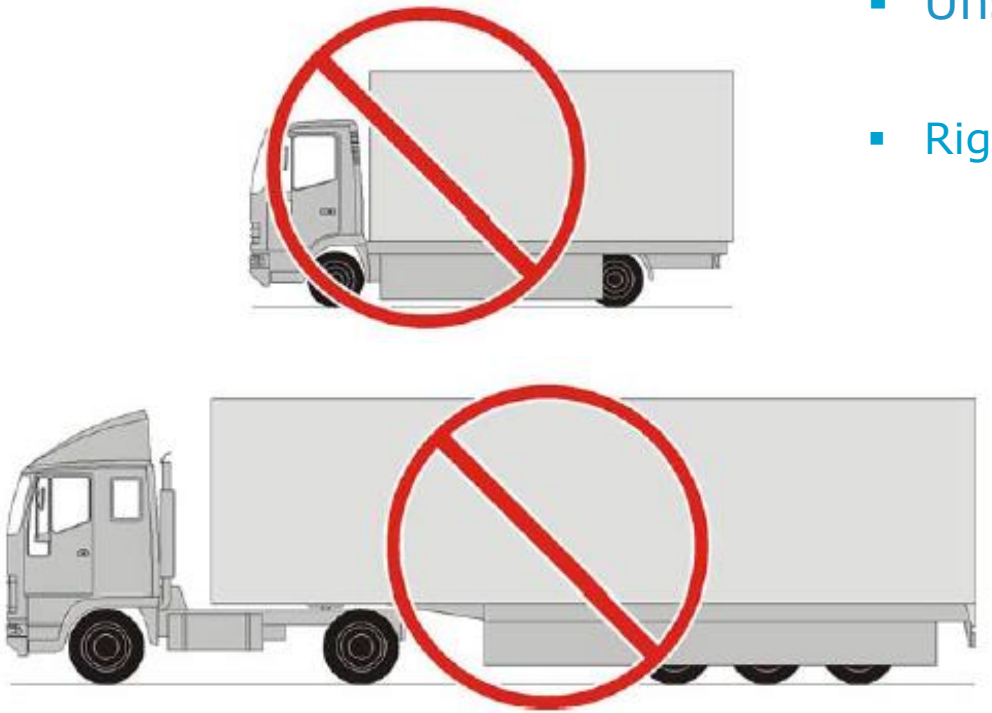
- Flashing RED and Beeper Alarm during the movement
- Blocking arm retracts and carriage move back
- Attention!
  This action is only possible when the dock leveler is in parking position and the sectional door is closed!

DE6090AR Automatic Restraint



- Free space in front of the wheel for the blocking arm

DE6090AR Automatic Restraint



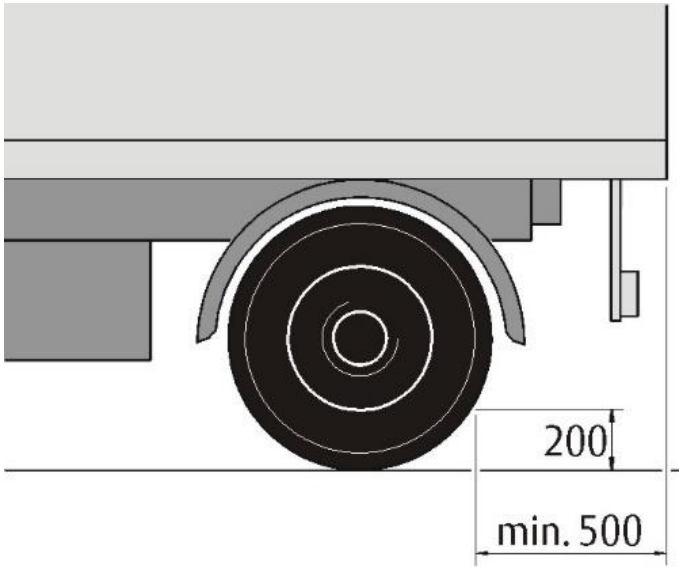
- Unsuitable Trucks
- Rigid covers of the wheels

DE6090AR Automatic Restraint

- Unsuitable trucks

  Ask if the customer has trucks with containers!

  The rear set of wheels must be min. 500 mm (in 200 mm height) in front of the buffer

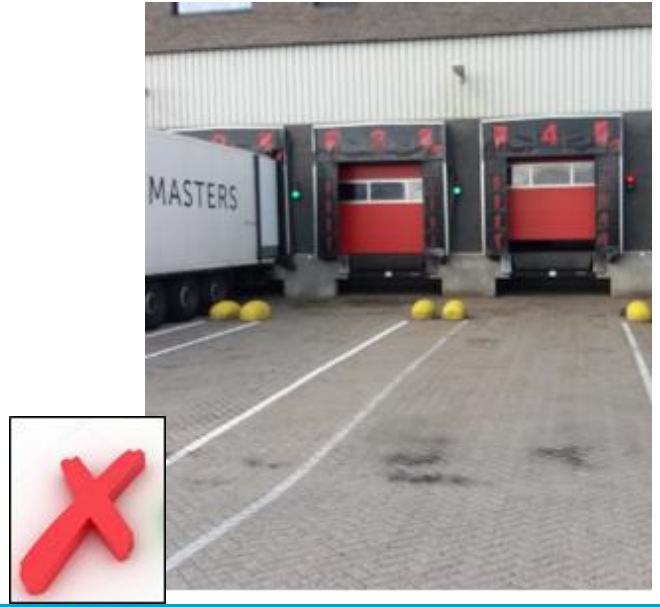


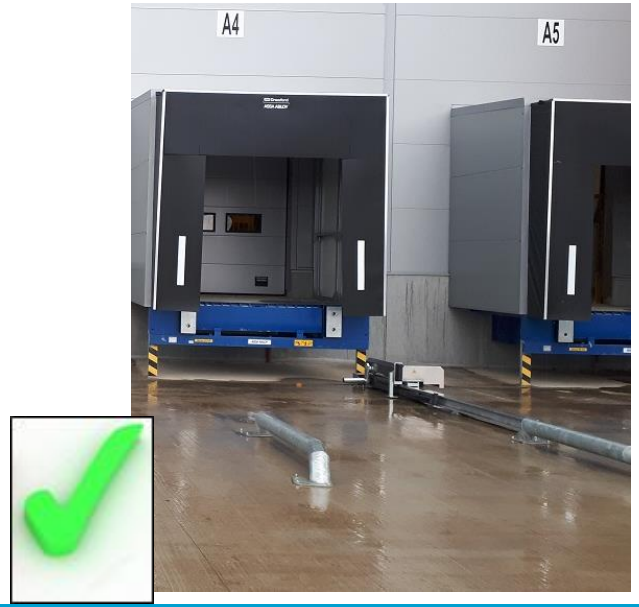


DE6090AR Automatic Restraint

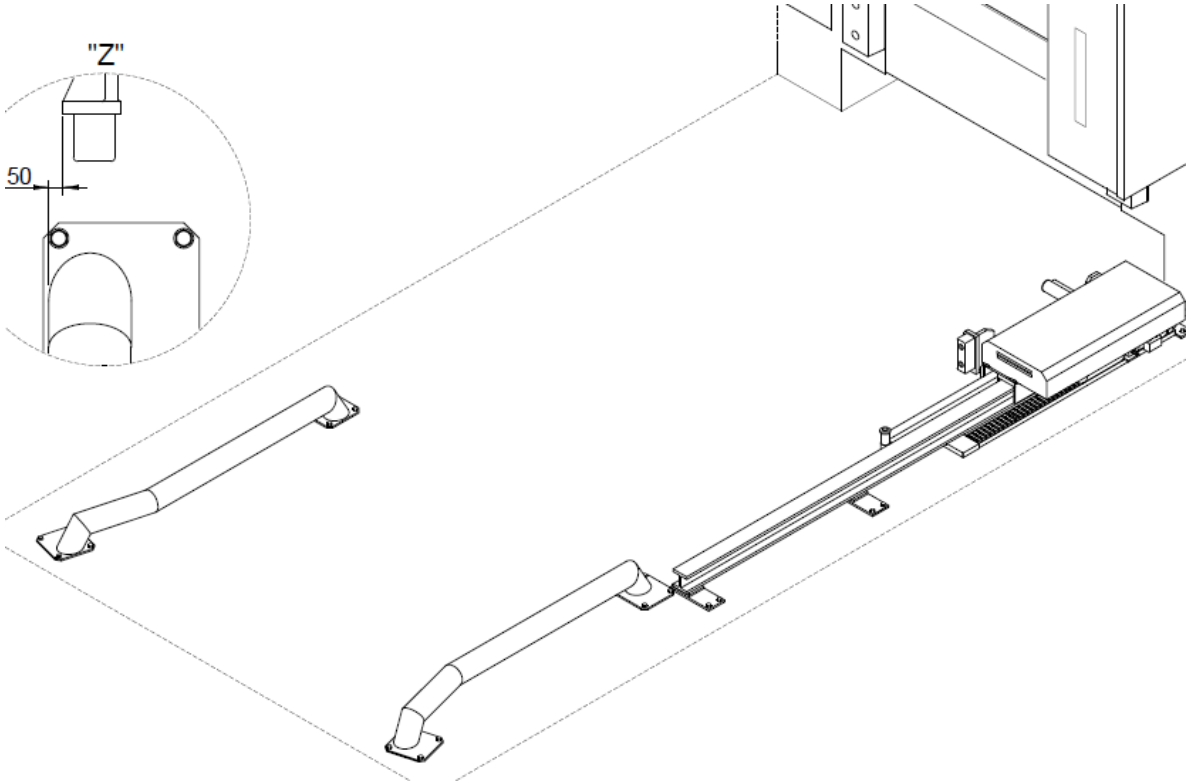
- Requires minimal building preparation

- BUT the Driving Surface on the Yard must be strong enough!. Existing Yard with Bricks will not be strong enough. Follow the Concrete Specification: C20/25 (B25, 350 kg/m³)





DE6090AR Automatic Restraint



- Parking Guides in front of the restraint system!
- Total length with parking guides about 7000 mm
- Yard slope up to 5%



DE6090MR Manual Restraint System Introduction of the new product

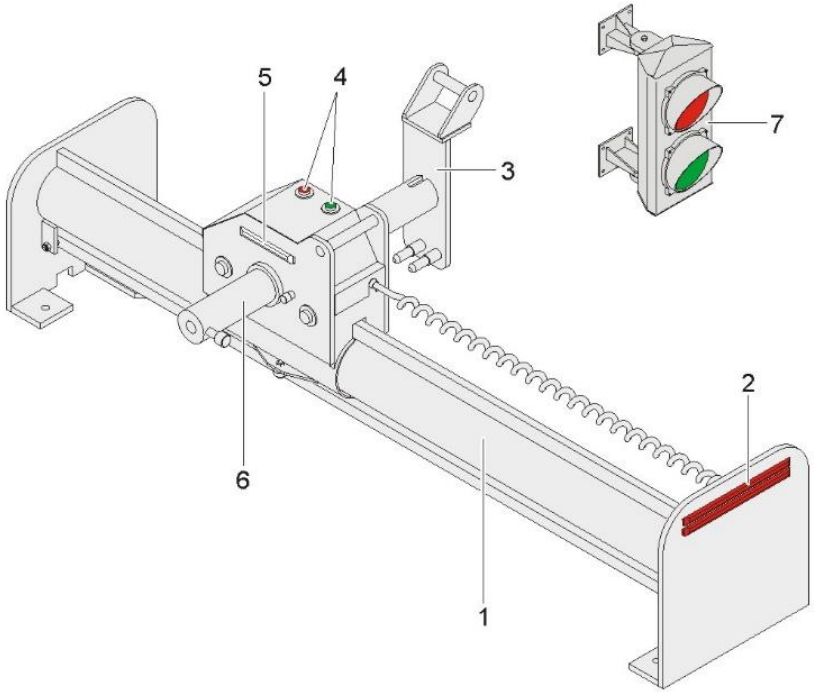
November 2018

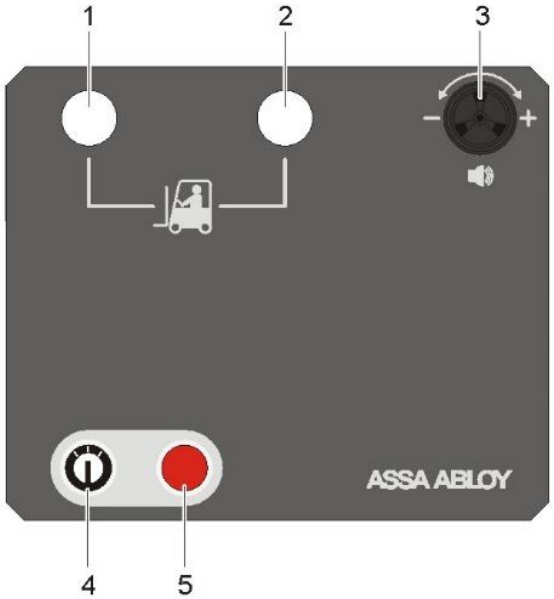
DE6090MR Manual Restraint



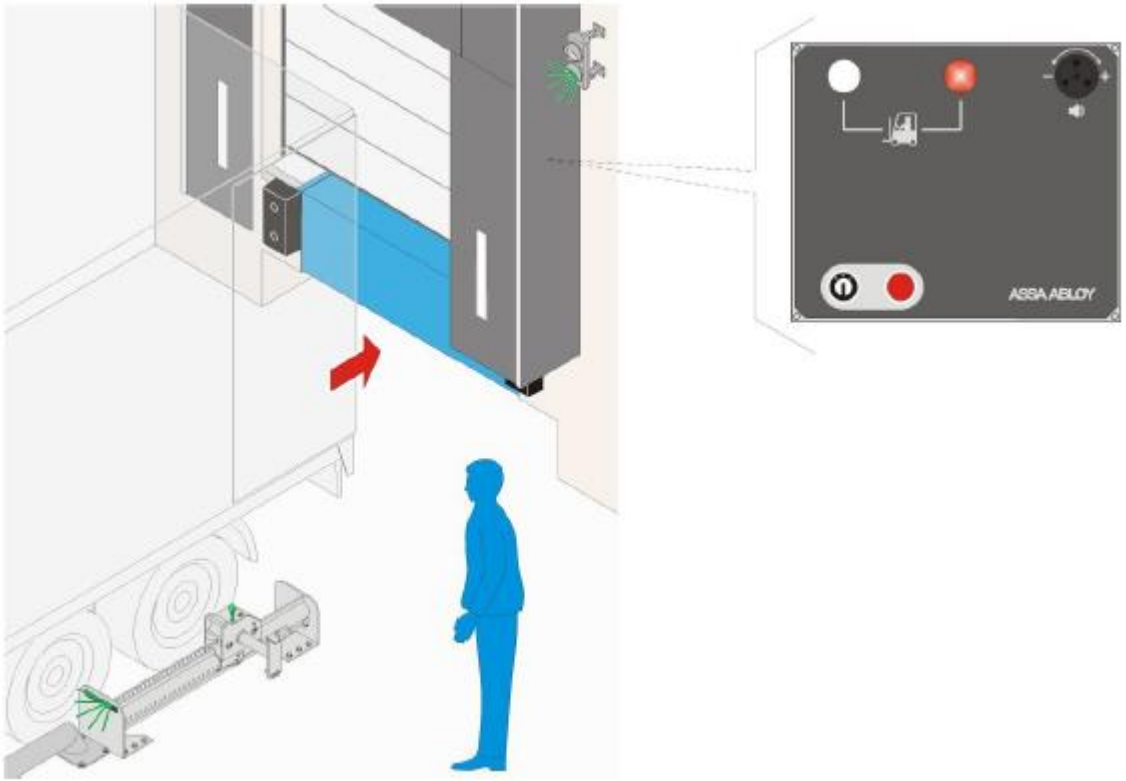


DE6090MR Manual Restraint





DE6090MR Manual Restraint

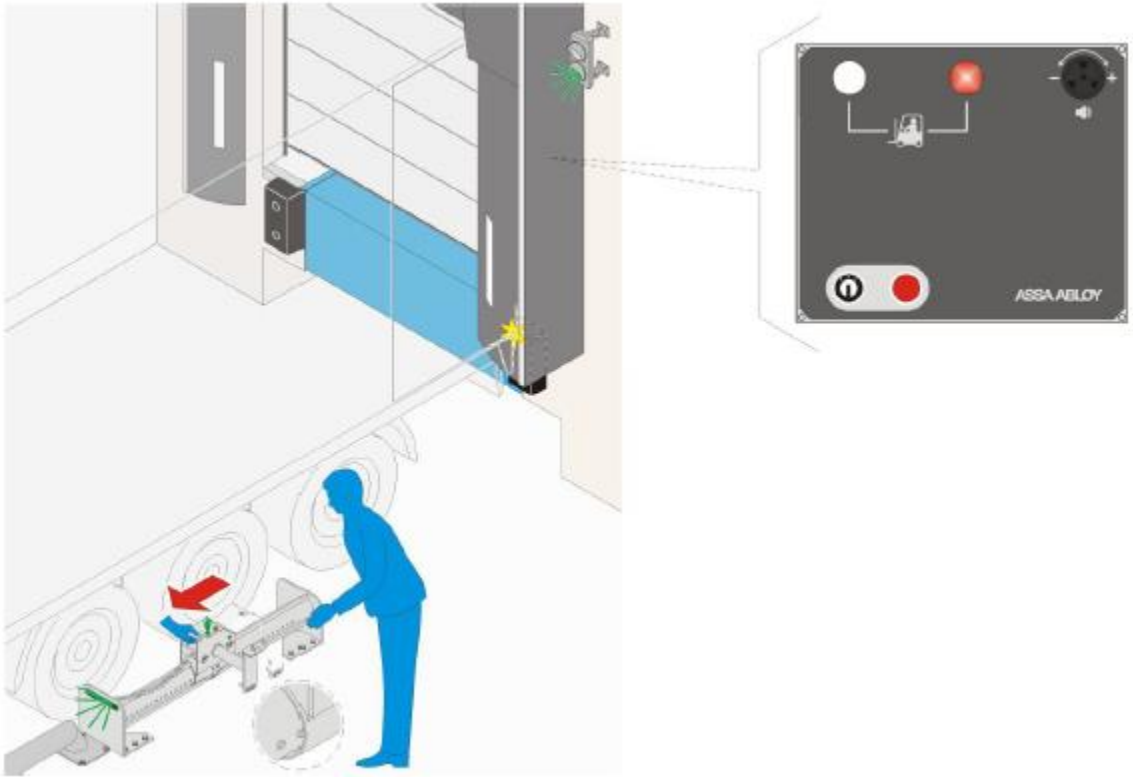


- Truck arriving

- Outside GREEN
- Signal Blocking Arm GREEN

- Inside RED
- The door is blocked

DE6090MR Manual Restraint

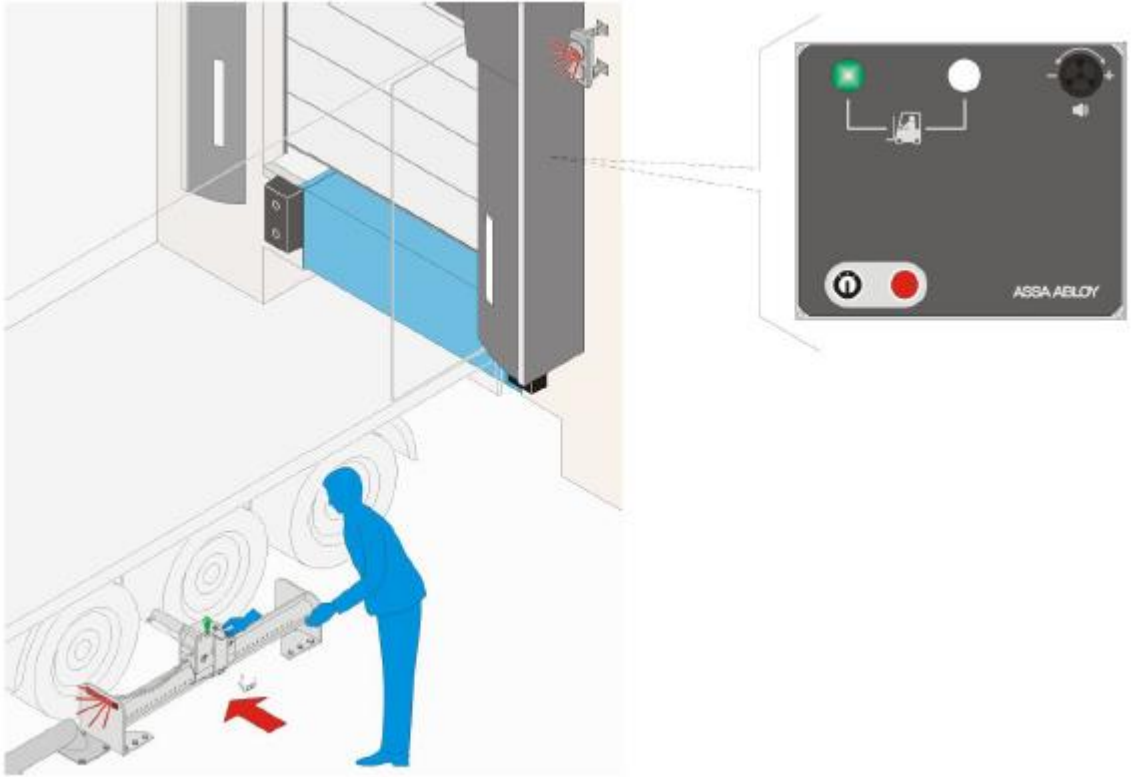


- Truck in place
- Positioning the Blocking Arm
- Must not be the most rear Wheel

- Outside GREEN
- Signal Blocking Arm GREEN

- Inside RED
- The door is blocked

DE6090MR Manual Restraint

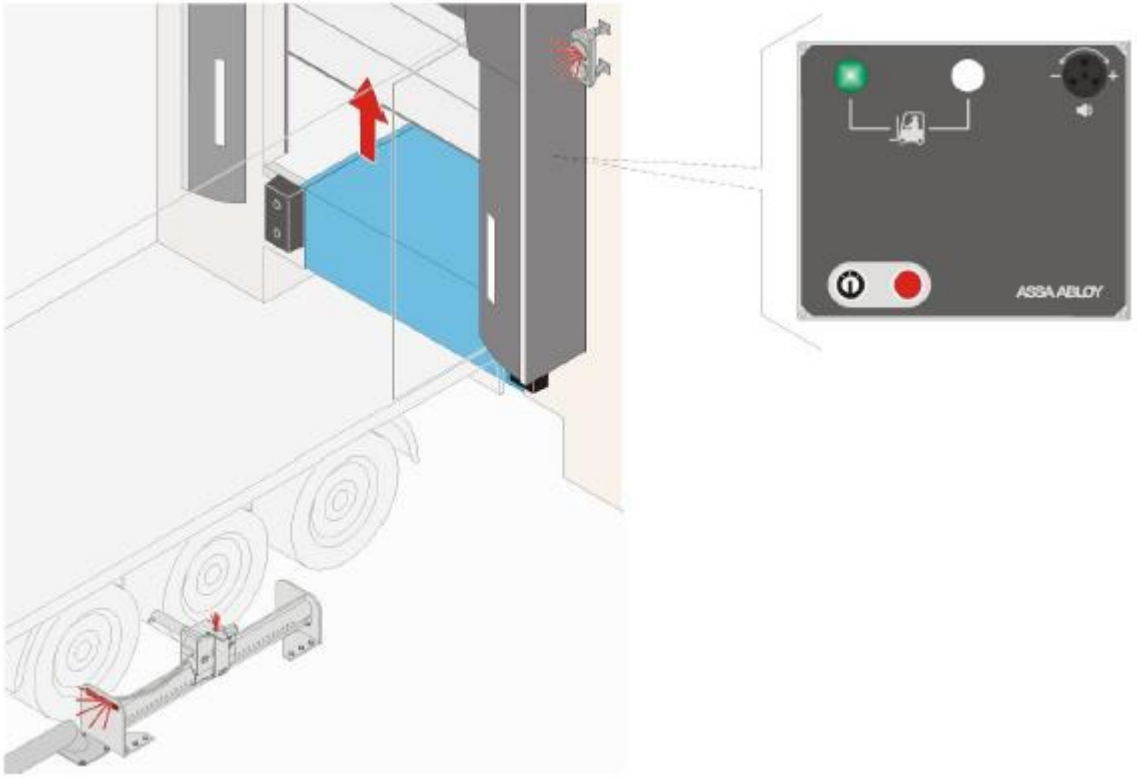


- Push Blocking Arm FORWARD

- Outside ROT
- Signal Blocking Arm GREEN
- The Arm is not blocked

- Inside GREEN
- The Door is released for use

DE6090MR Manual Restraint

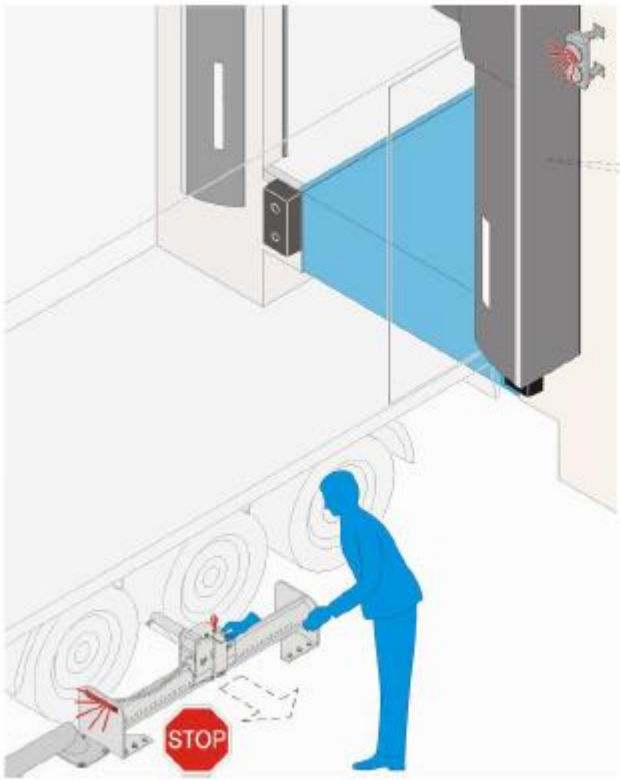


- Open the Door
- Loading operation

- Outside RED
- Signal Blocking Arm RED
- The Arm is blocked (Electro magnetic)

- Inside GREEN

DE6090MR Manual Restraint



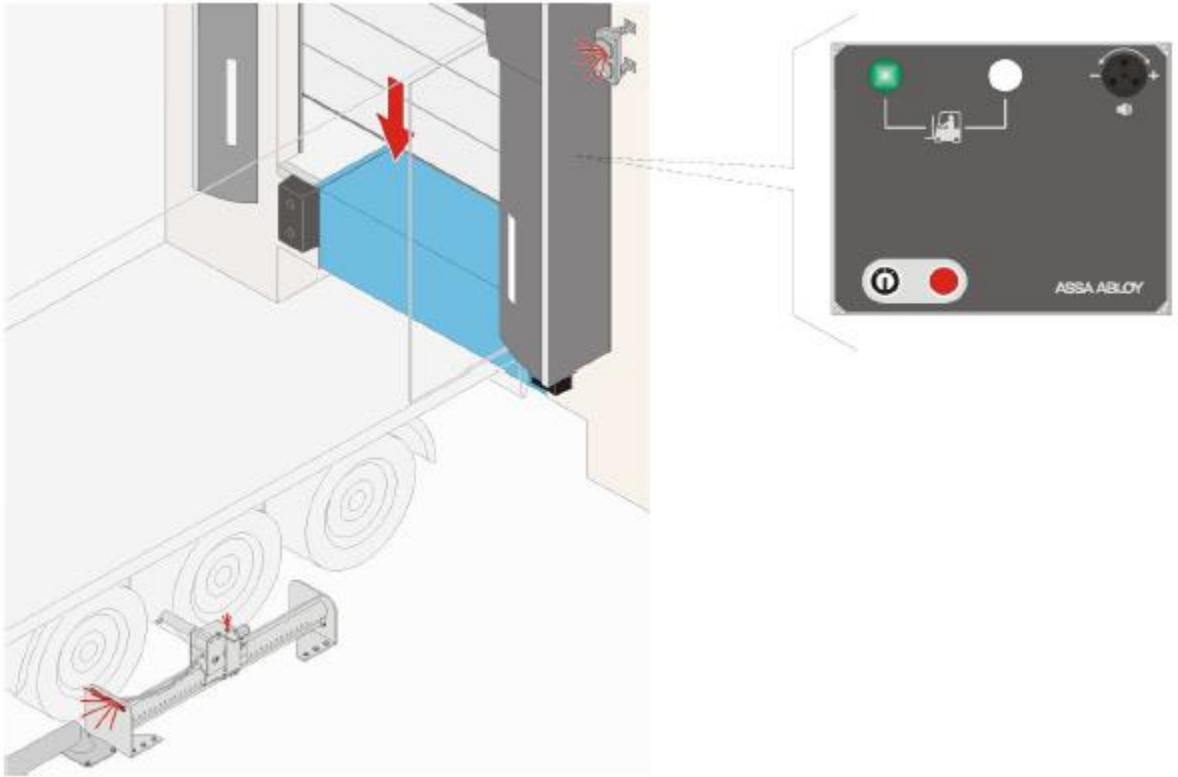


- Try to remove the Blocking Arm (is NOT possible)
- Beeper Alarm sounds

- Outside RED
- Signal Blocking Arm RED

- Inside GREEN

DE6090MR Manual Restraint

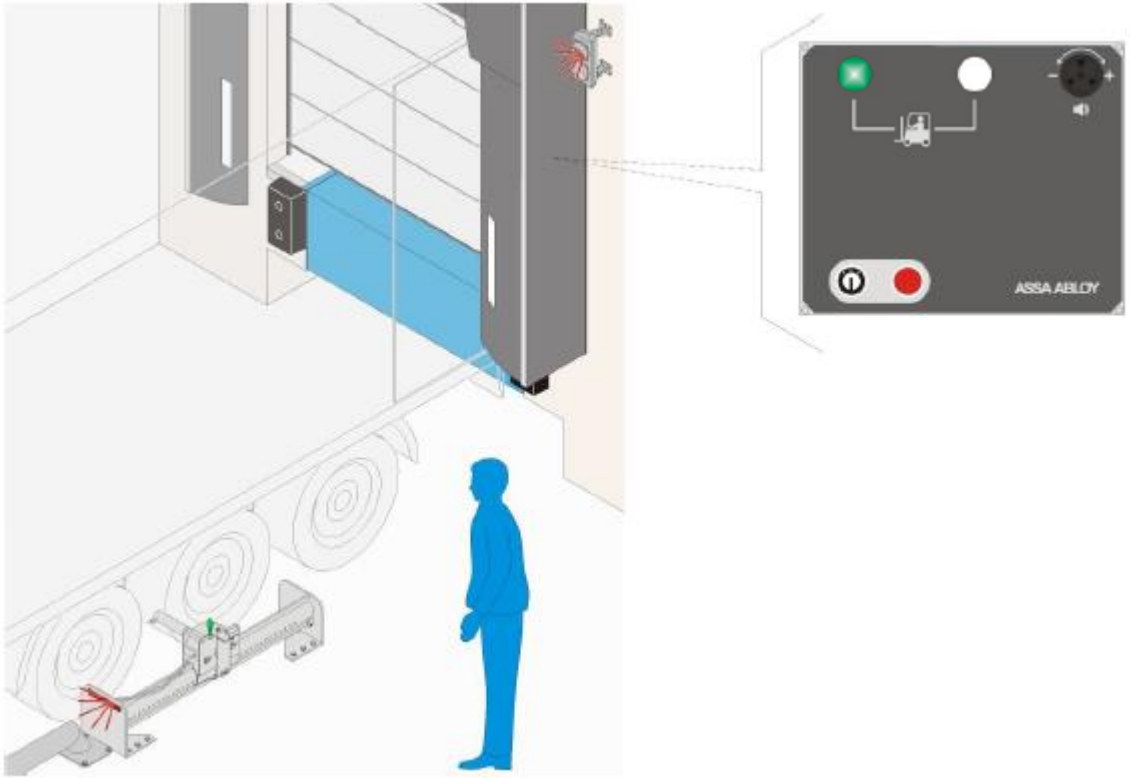


- End of loading
- Door going down

- Outside RED
- Signal Blocking Arm RED
- The Arm is blocked

- Inside GREEN

DE6090MR Manual Restraint



- Door CLOSED

- Outside GREEN
- Signal Blocking Arm GREEN
- The Arm is released for use

- Inside GREEN

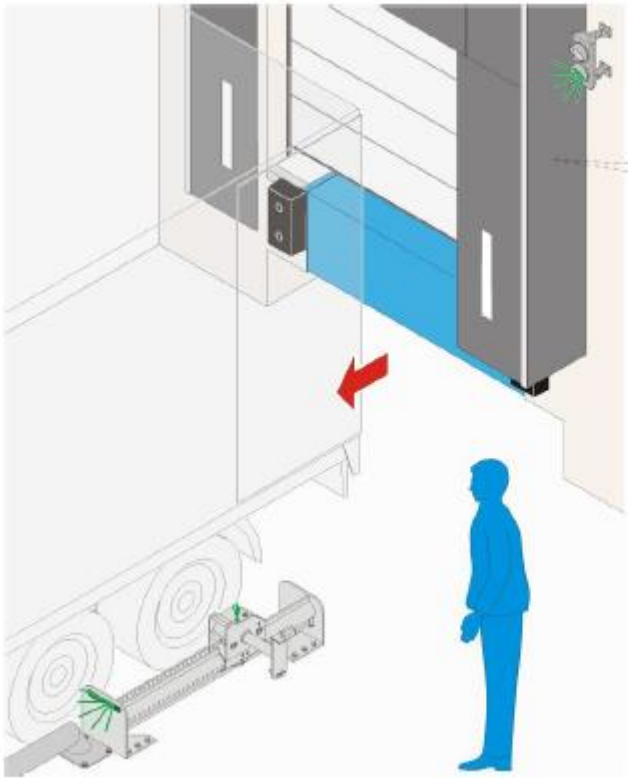
DE6090MR Manual Restraint





- Blocking BACK
- Outside GREEN
- Signal Blocking Arm GREEN
- Inside RED
- The Door is blocked

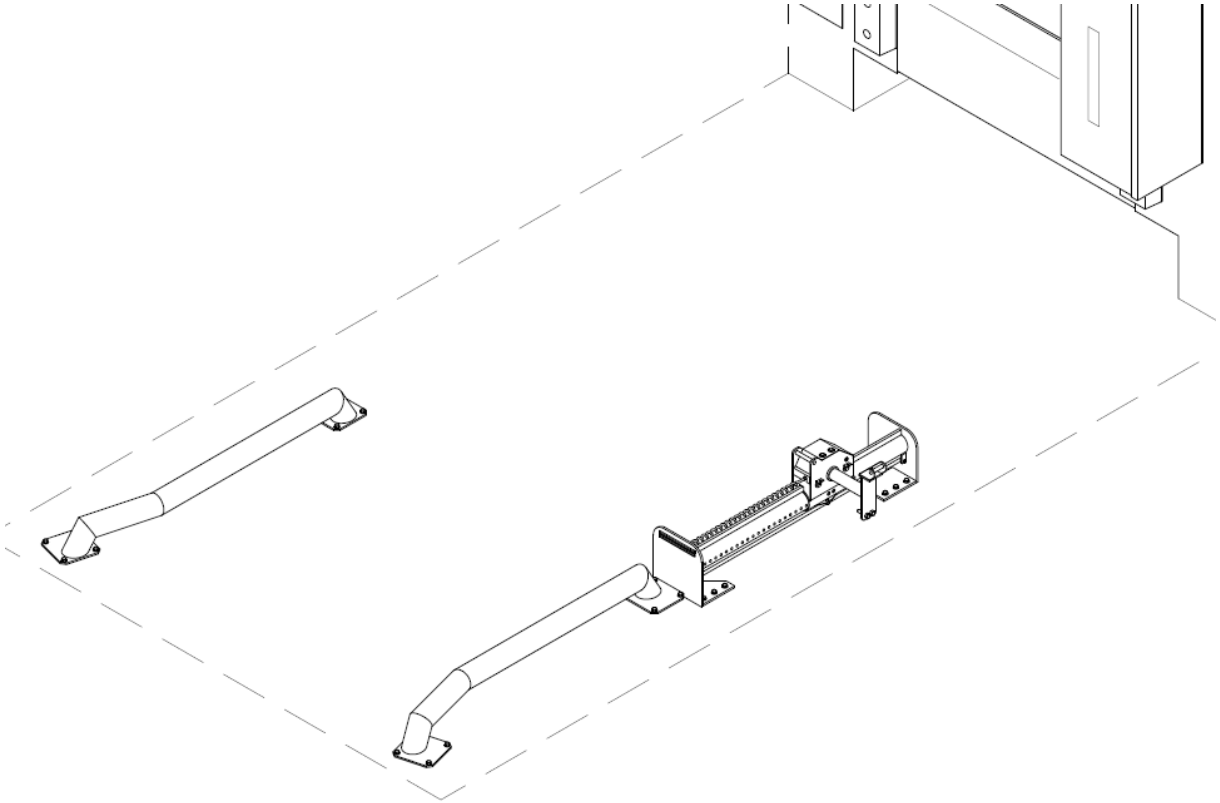
DE6090MR Manual Restraint





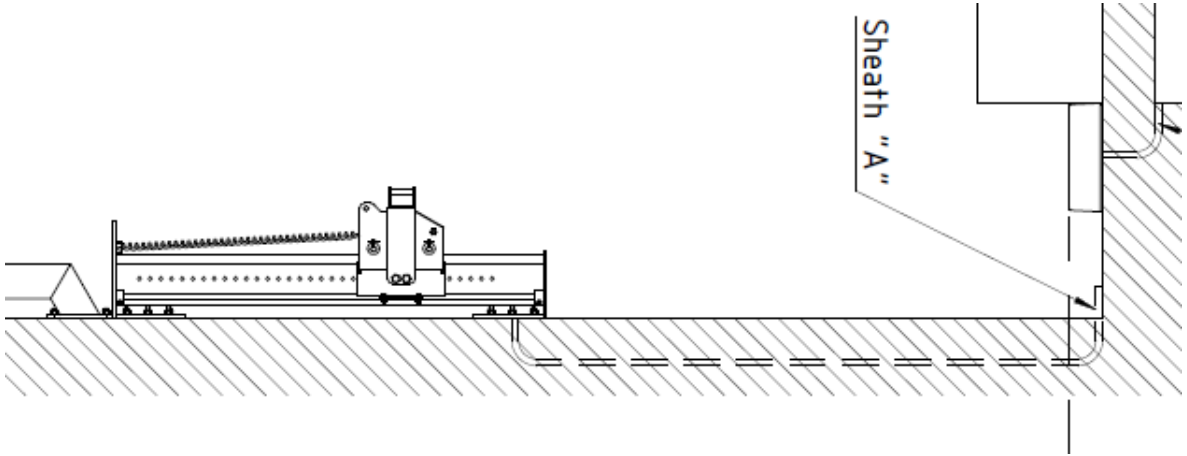
- Truck driving OFF
- Outside GREEN
- Signal Blocking Arm GREEN
- Inside RED
- The Door is blocked

DE6090MR Manual Restraint



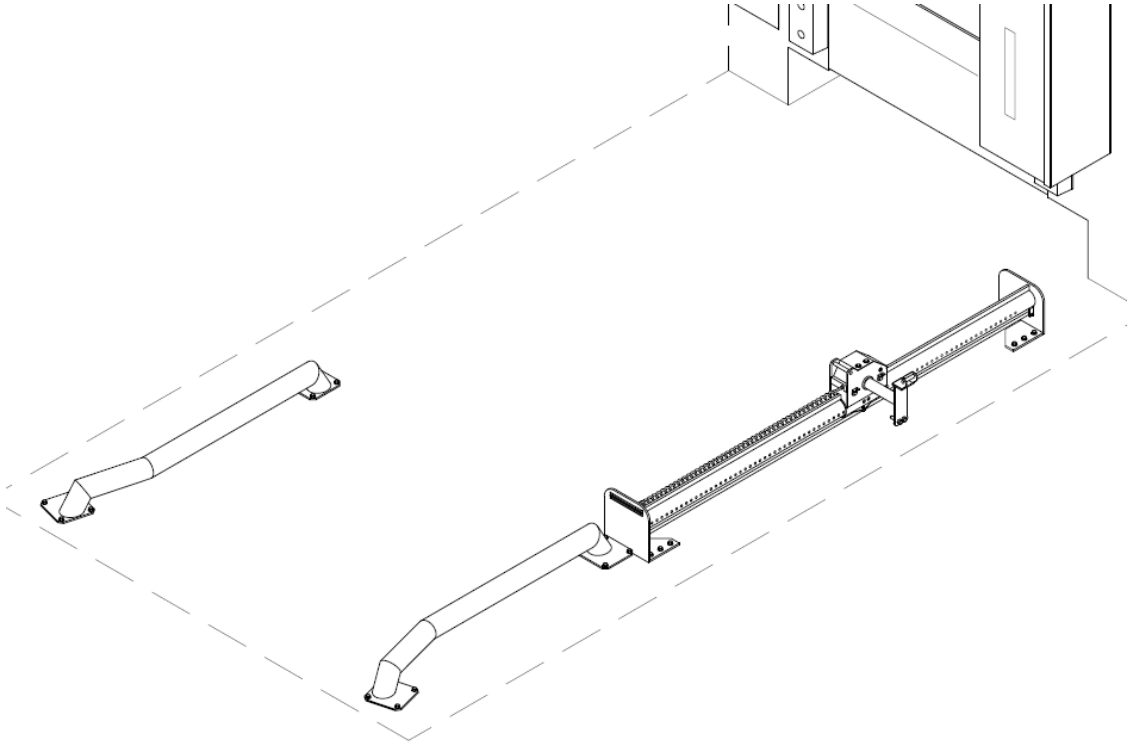
- Standard model H50
- 2800 mm from the building
- Movable operational stroke 1800 mm
- Parking guides in front of the System

DE6090MR Manual Restraint



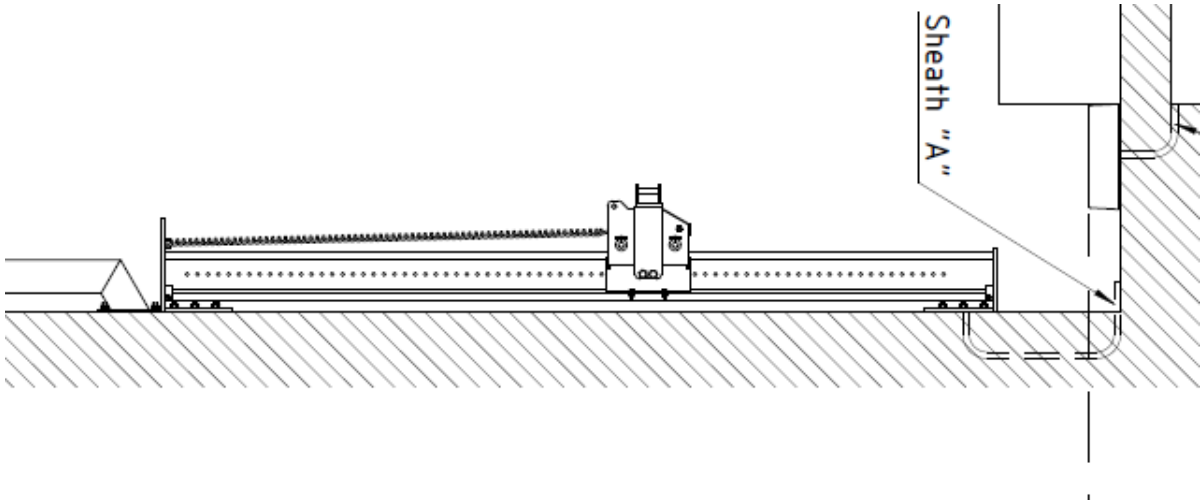
- Standard model H50
- 2800 mm from the building
- Pipe for cables as part of building preparation by others
- MUST be below ground!

DE6090MR Manual Restraint



- Standard model H50
- 800 mm from the building
- Movable operational stroke 3800 mm
- Parking guides in front of the System

DE6090MR Manual Restraint



- Standard model H50
- 800 mm from the building
- Pipe for cables as part of building preparation by others
- MUST be below ground!


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